

GENX: Porewater SO₄/Cl

May 2026 Samples

2026-09-09

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-09-02" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "May"
Site = "GCREW"
run_notes = "CH2-80 Cl is very high (321mM) with a bad spike,
            likely human error, will re-run with GENX July run" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/GENX_202605_Cl.txt"
raw_file_name_so4 = "Raw Data/GENX_202605_SO4.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCReW_GENX_PW_Processed_Cl_SO4_202605.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCReW", "GCReW_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2025.csv"
```

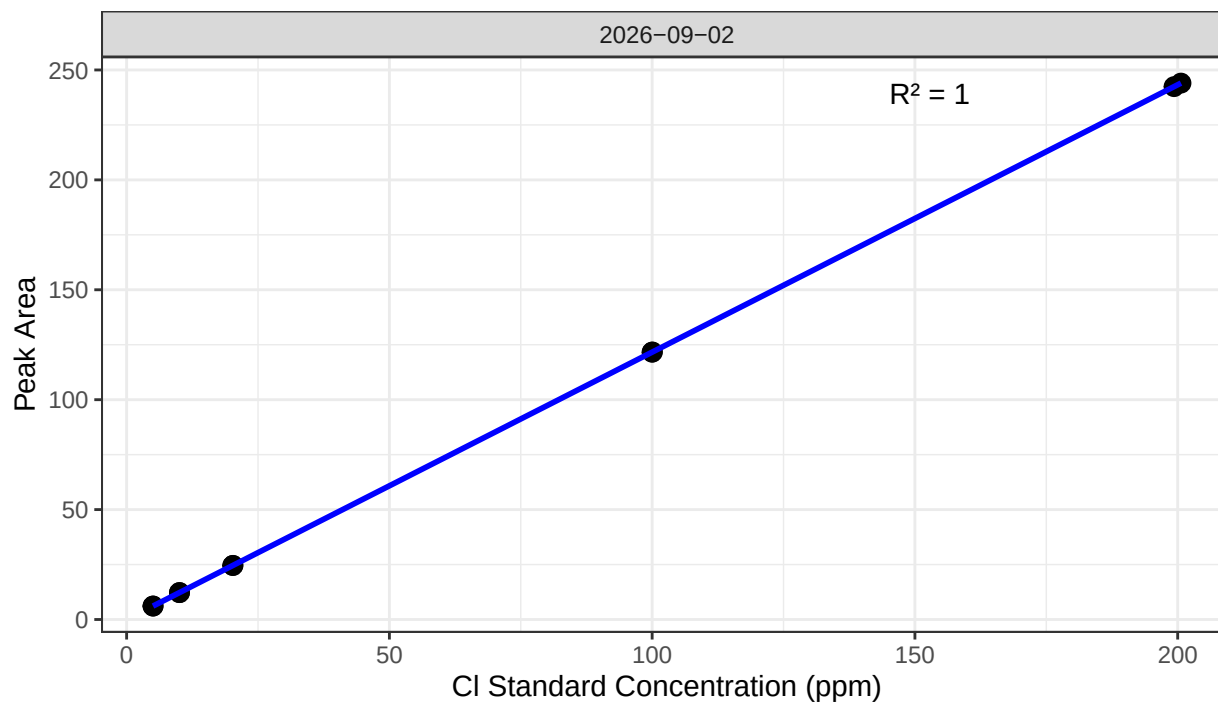
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

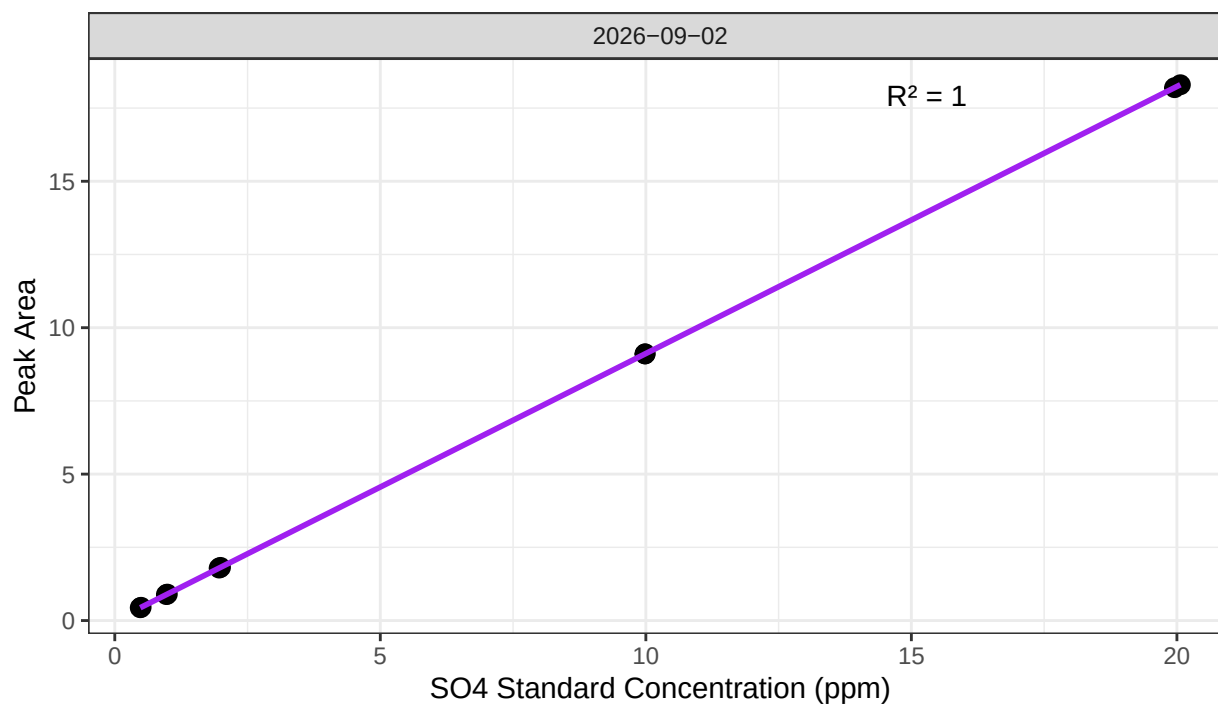
##Import Sample Data

0.2 Assess Standard Curves

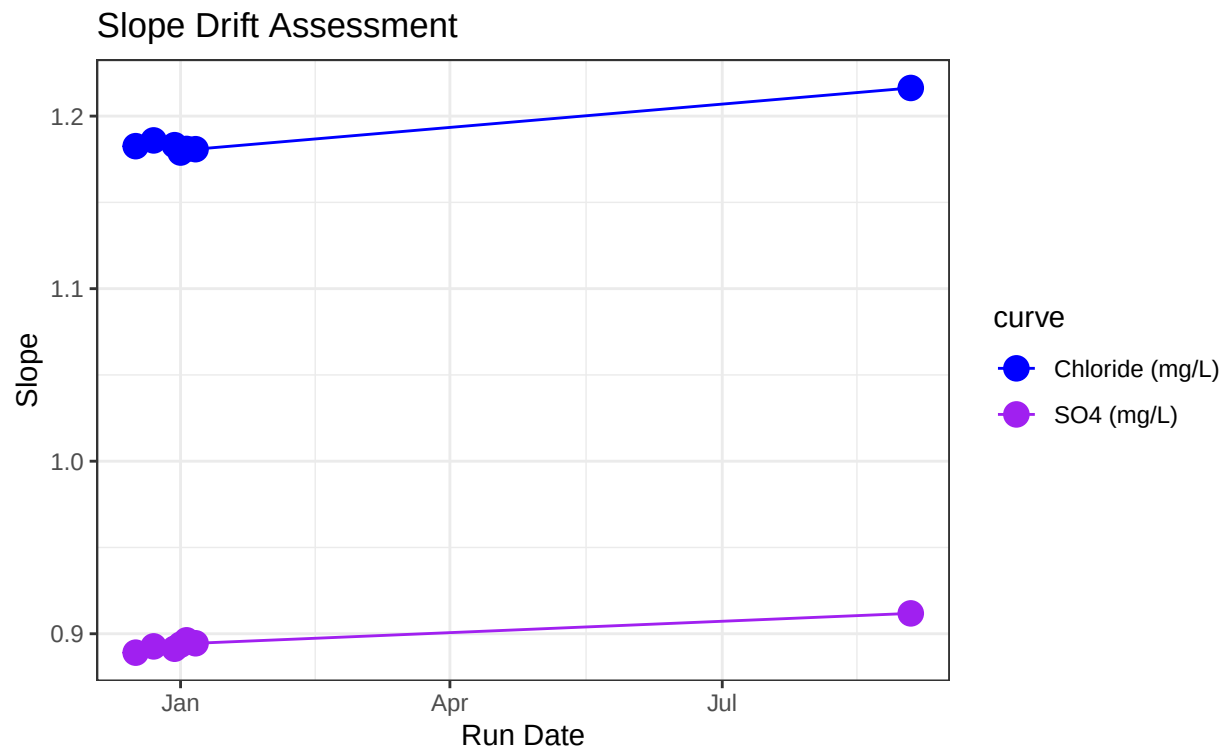
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

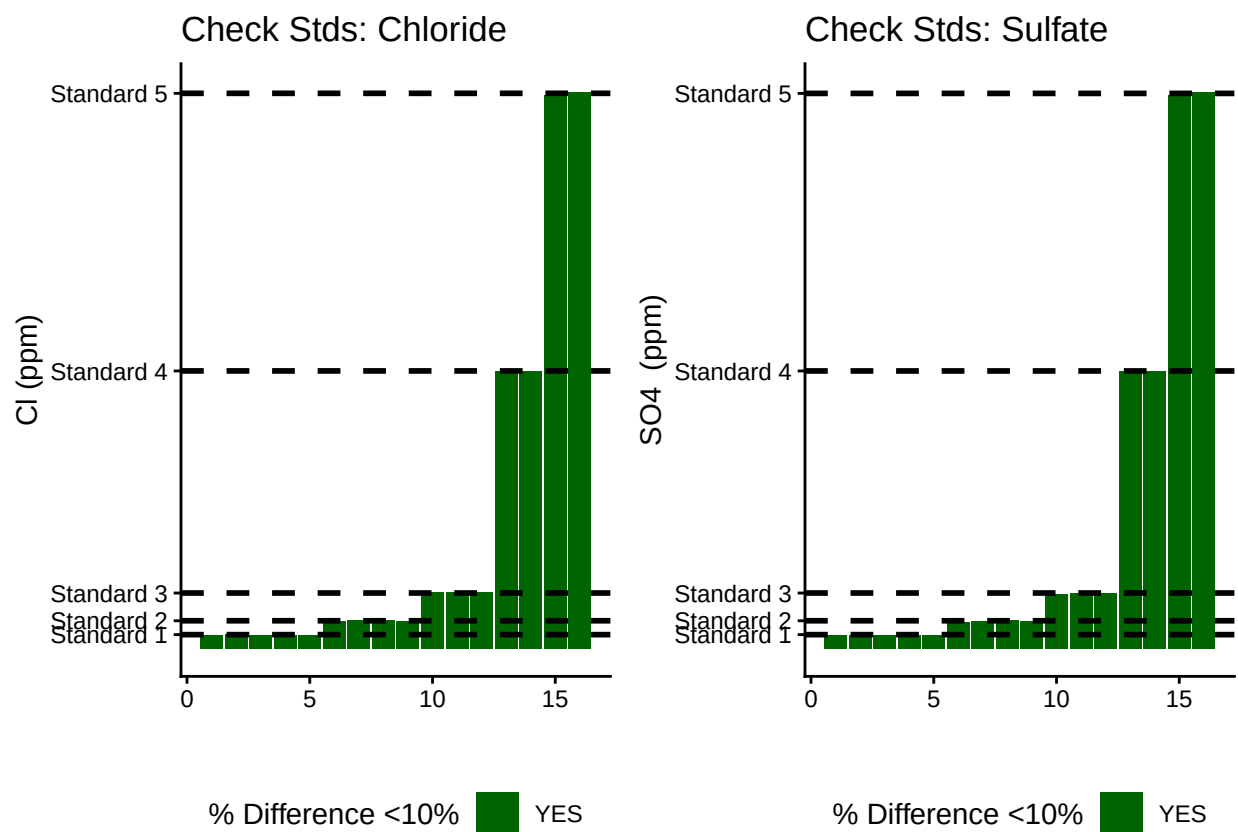
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 5.03 0.0165 0.00327 Chloride Check Standard RSD within Range --
## 2 Standard 2 10.1 0.0732 0.00724 Chloride Check Standard RSD within Range --
## 3 Standard 3 20.2 0.0246 0.00122 Chloride Check Standard RSD within Range --
## 4 Standard 4 100. 0.0486 0.000486 Chloride Check Standard RSD within Range --
## 5 Standard 5 200. 0.933 0.00467 Chloride Check Standard RSD within Range --
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1 0.485 0.00623 0.0128 Sulfate Check Standard RSD within Range ~
## 2 Standard 2 0.991 0.0162 0.0164 Sulfate Check Standard RSD within Range ~
## 3 Standard 3 1.98 0.0167 0.00841 Sulfate Check Standard RSD within Range ~
## 4 Standard 4 9.99 0.00240 0.000241 Sulfate Check Standard RSD within Range ~
## 5 Standard 5 20.0 0.0774 0.00387 Sulfate Check Standard RSD within Range ~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



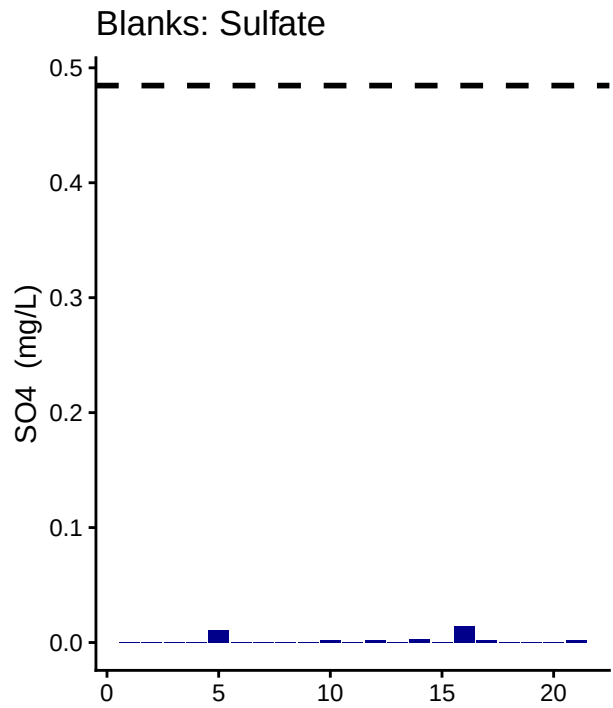
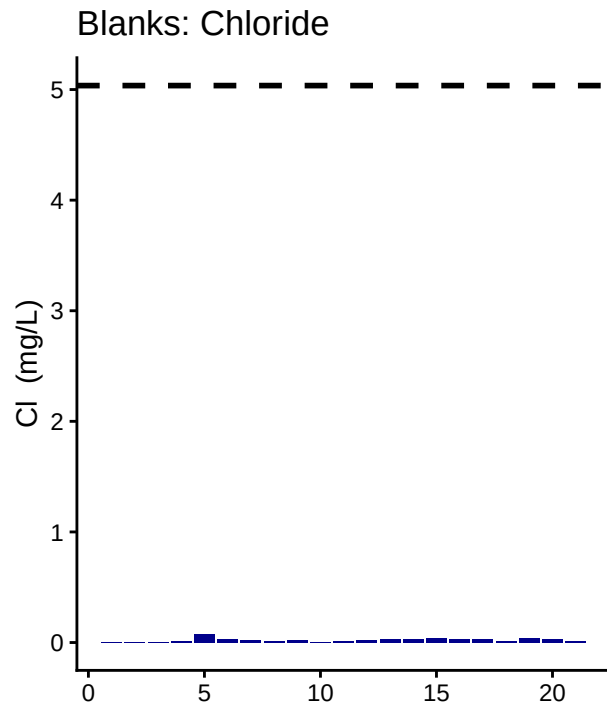
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples YE

Blank Conc <25% Quartile Samples YE

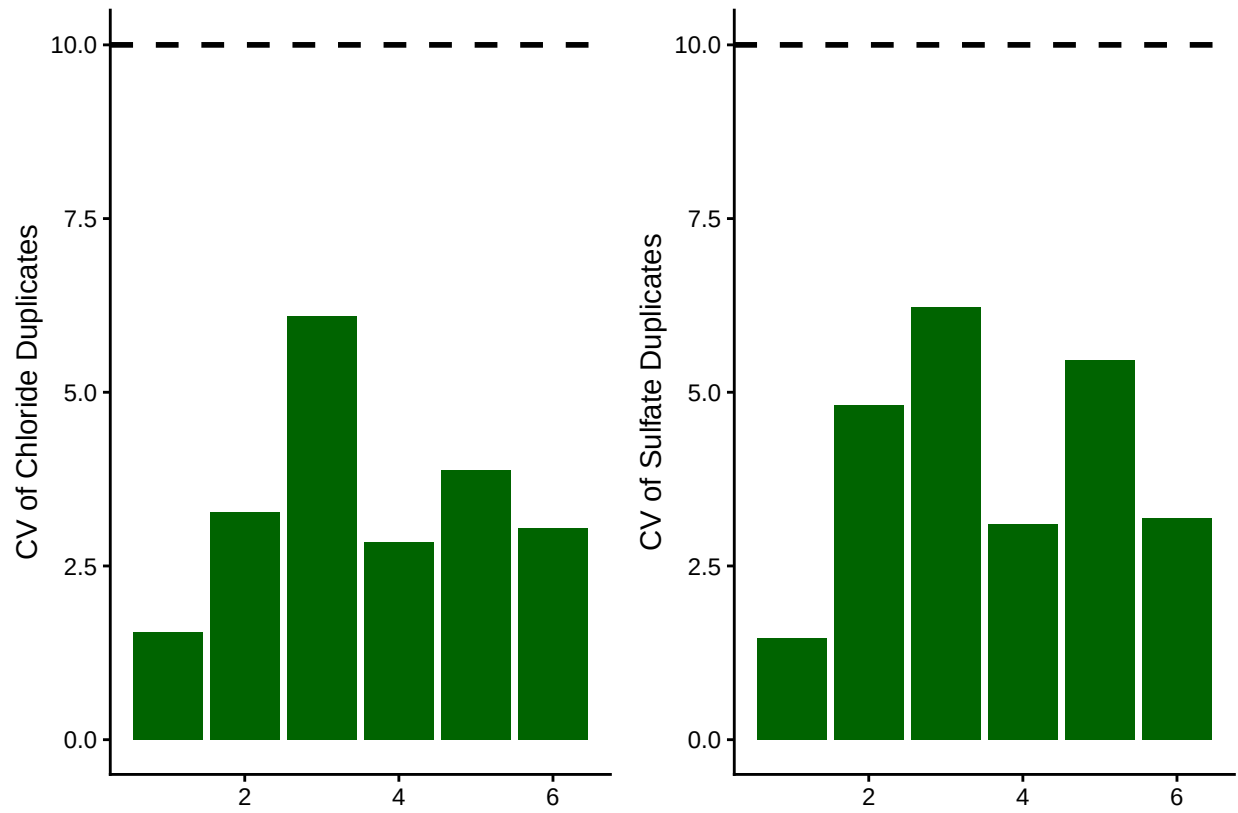
```
## Chloride blanks mean ppm:
```

```
## [1] 0.02336667
```

```
## Sulfate blanks mean ppm:
```

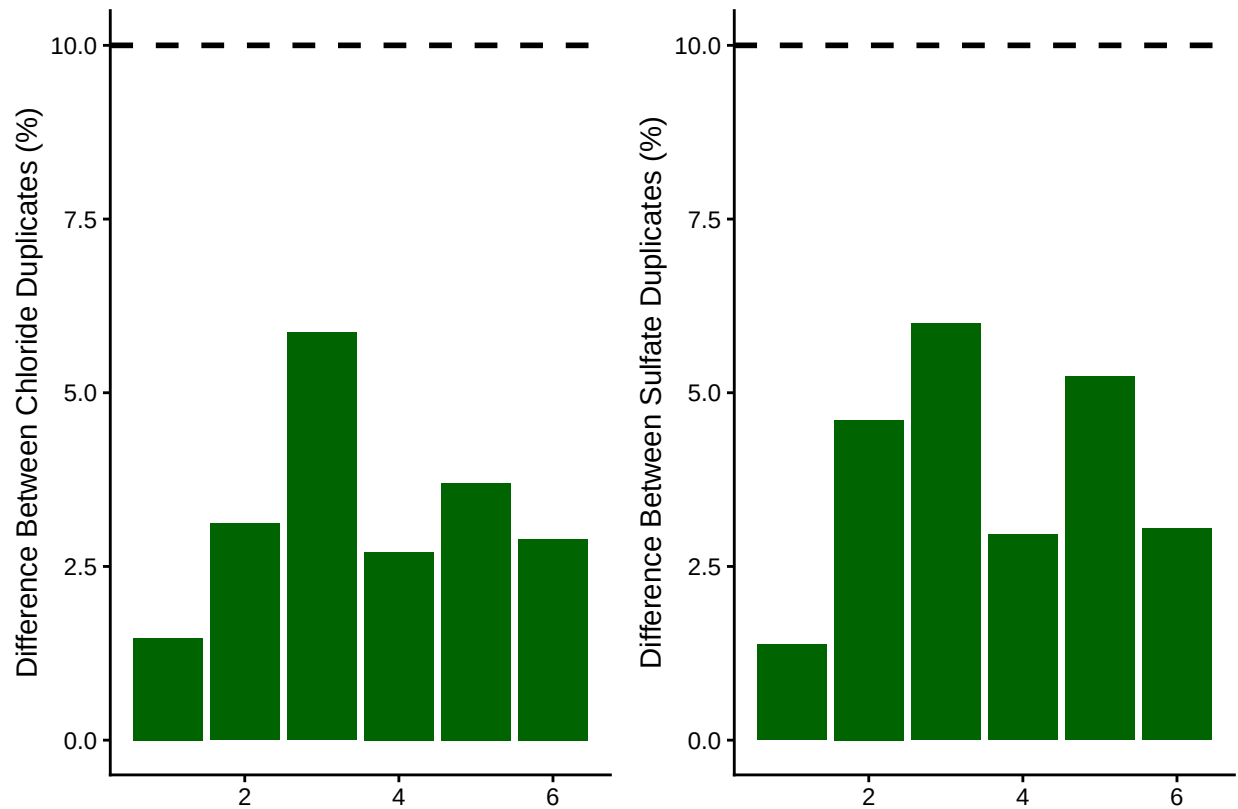
```
## [1] 0.001852381
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

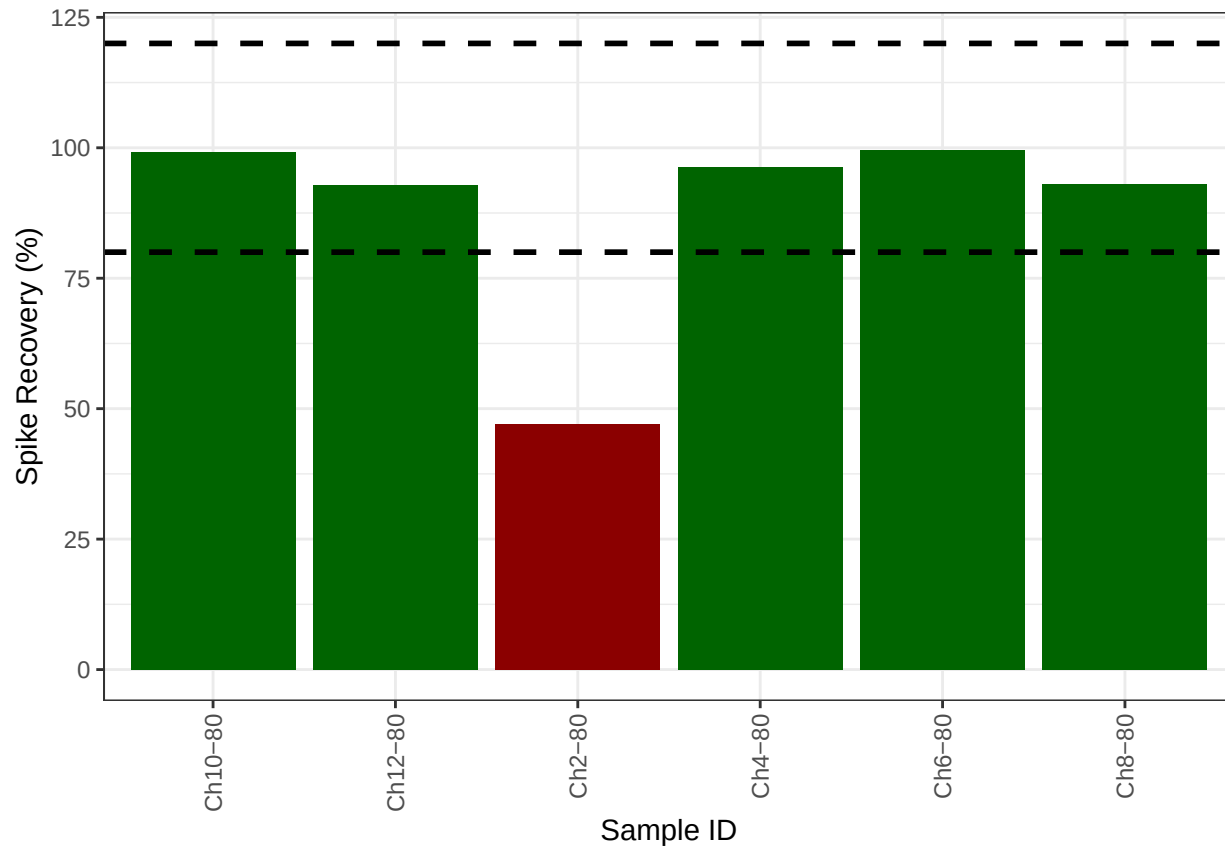
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
all_dat$Dilution <- Dilution
# head(all_dat)
```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

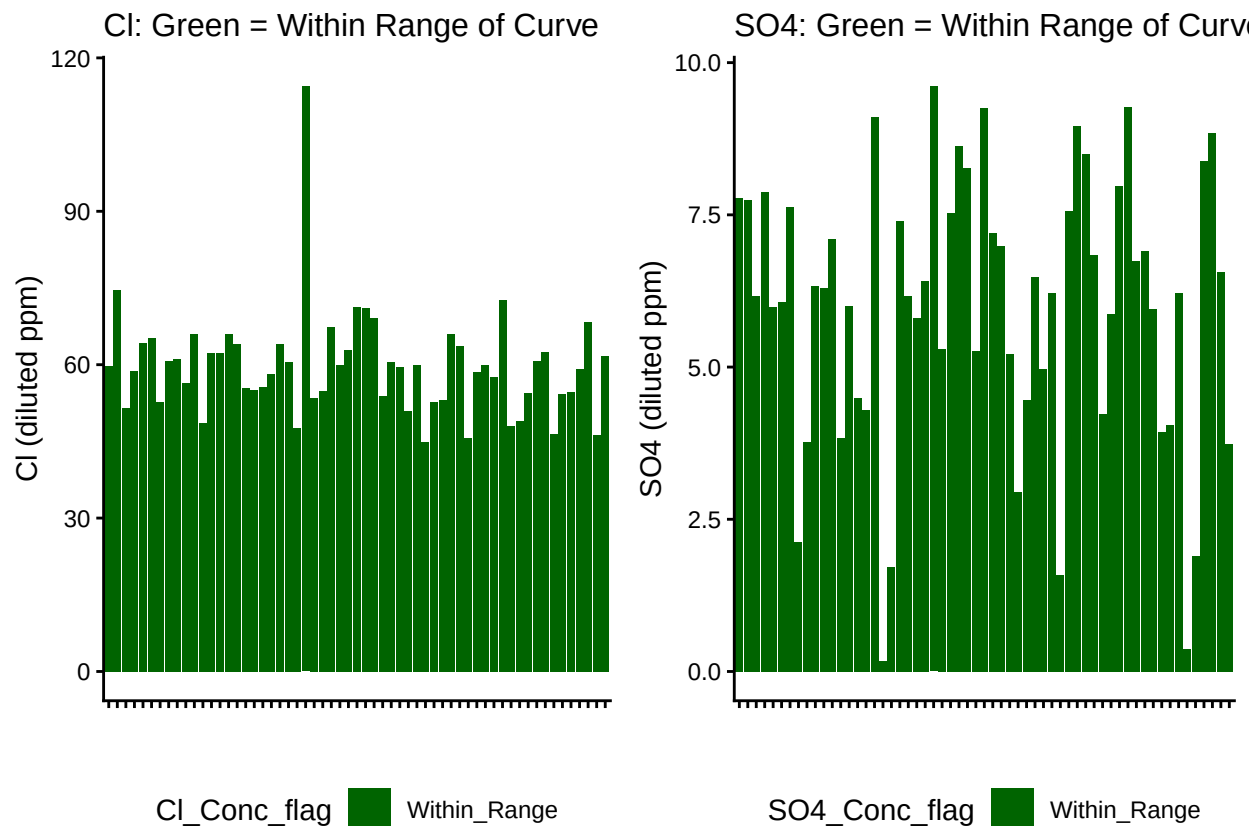


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

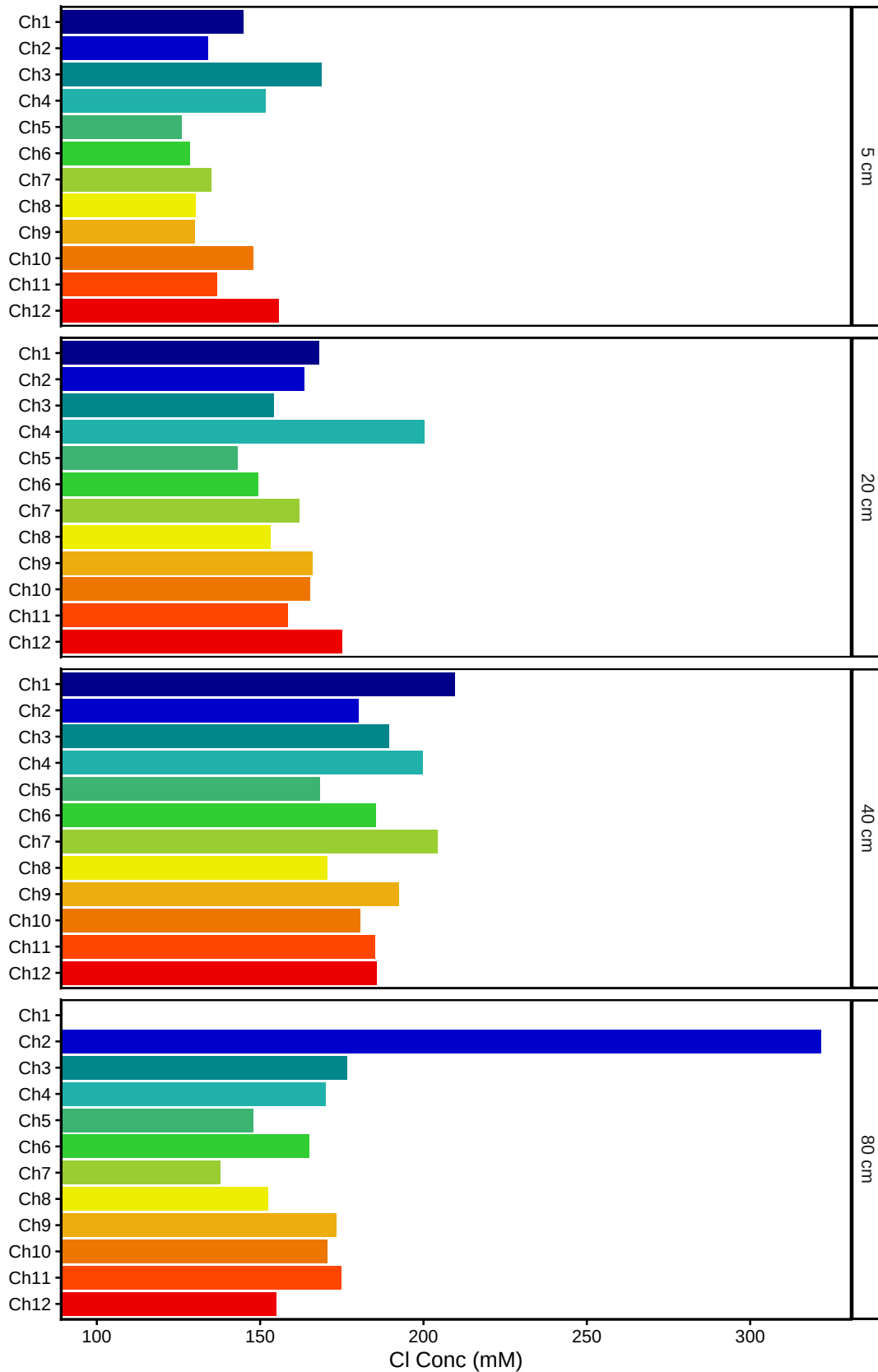
0.9 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```

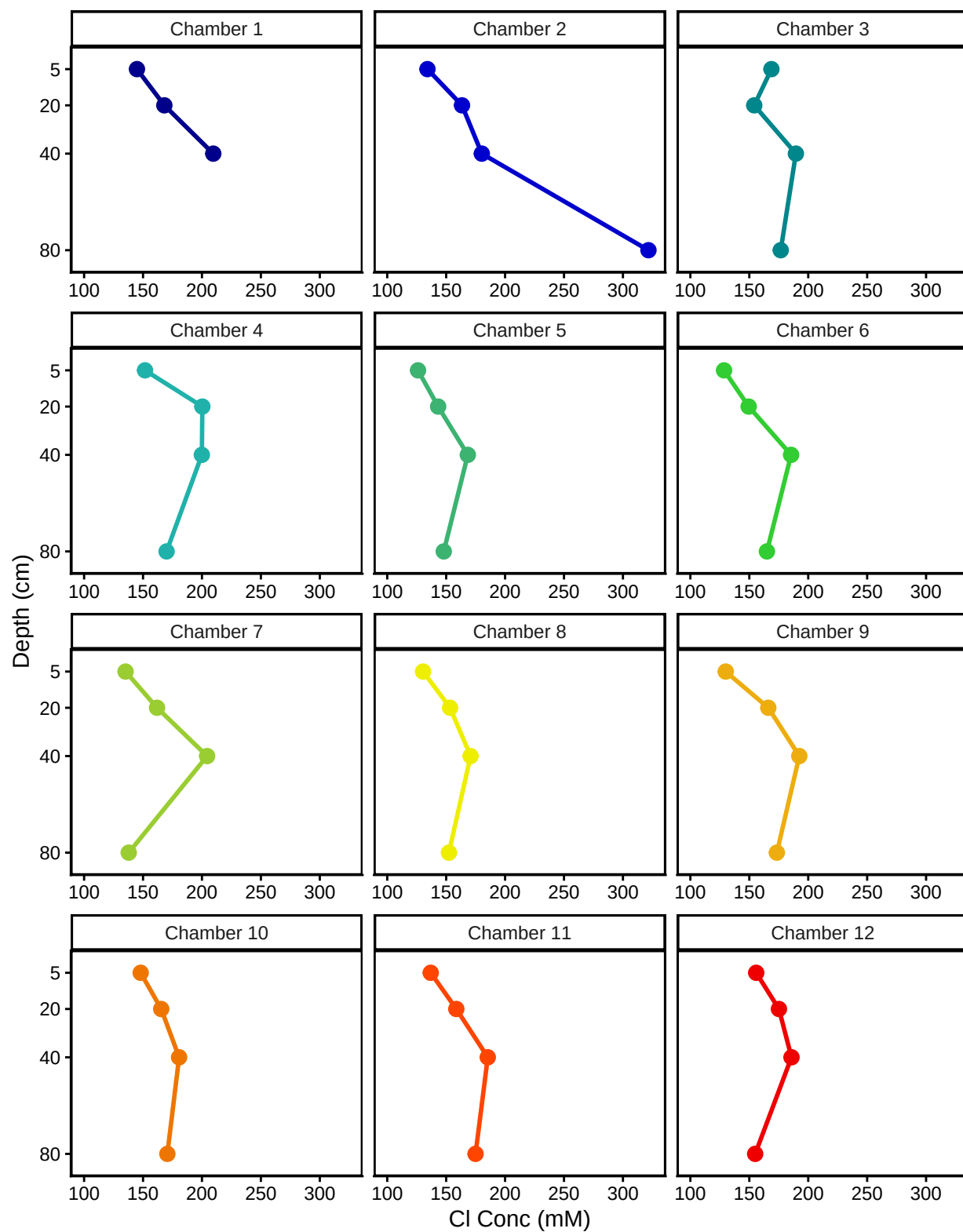
0.10 Visualize Data - Cl

```
## Visualize Data - Cl
```

GENX: Porewater Cl



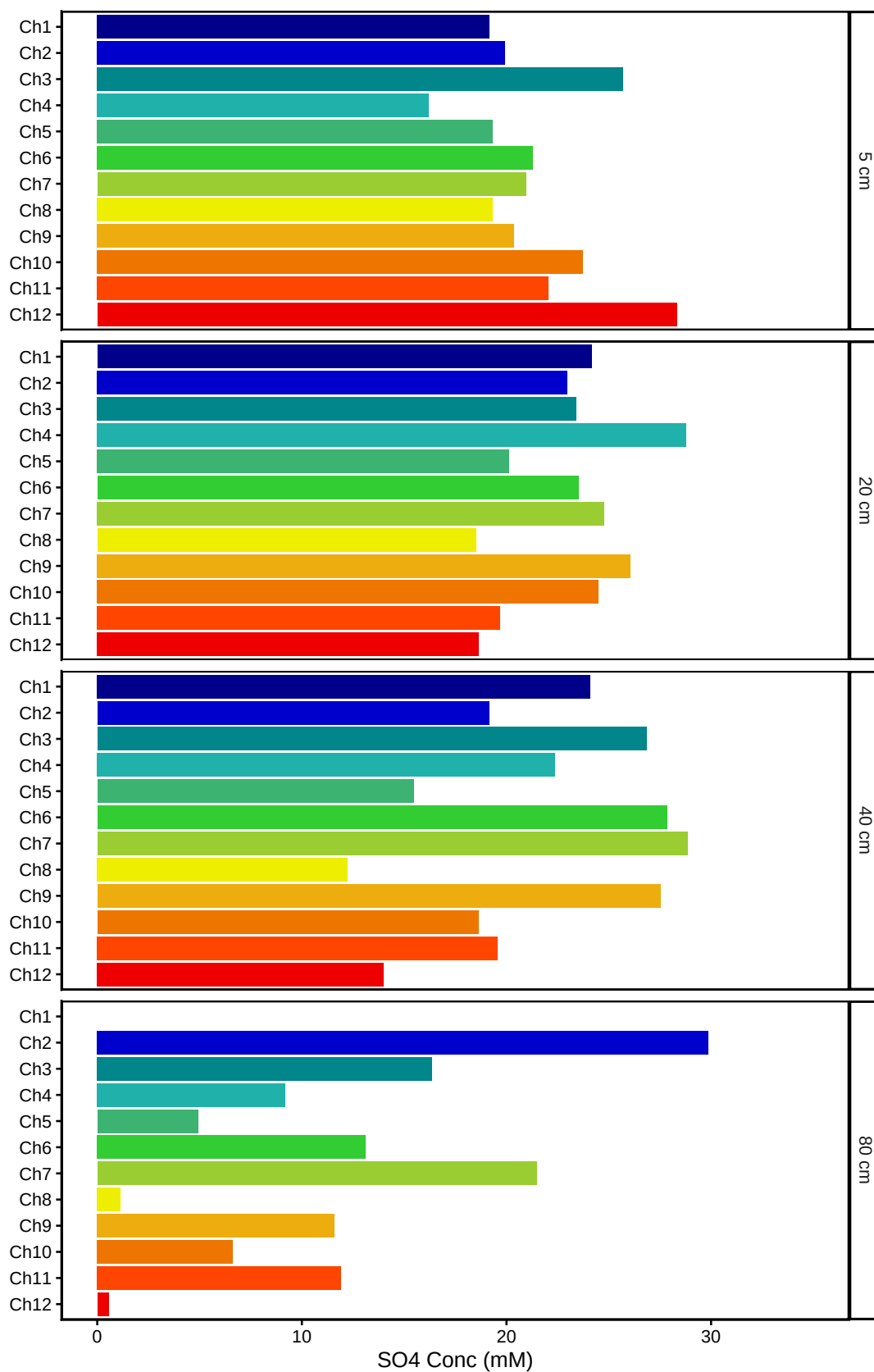
GENX: Porewater Cl by Chamber



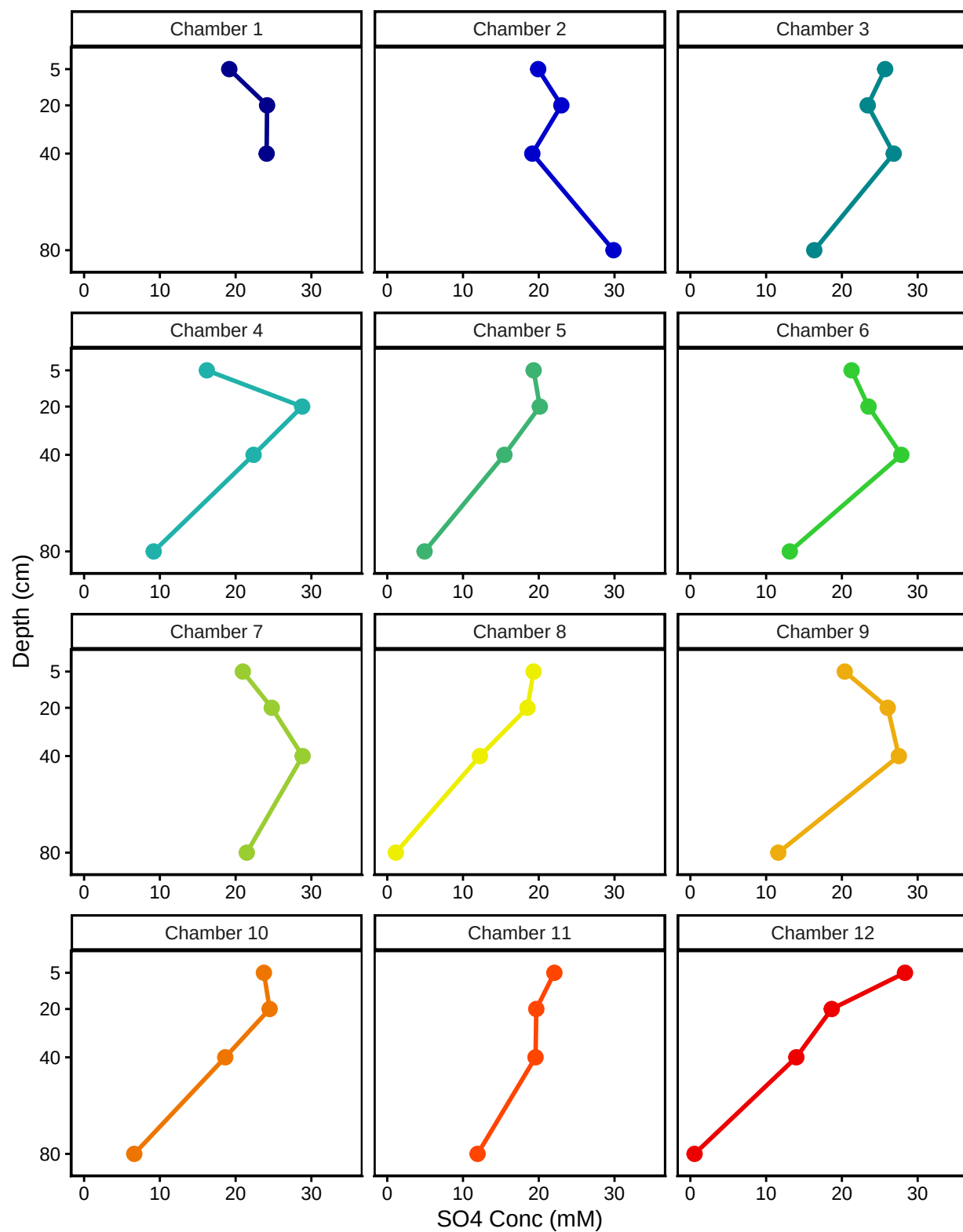
0.11 Visualize Data - SO4

```
## Visualize Data - SO4
```

GENX: Porewater SO4



GENX: Porewater SO₄ by Chamber



0.12 Export Processed Data

#end